

Logos Experiment 19

Quantization and Compression of the Experiment 18 Language Model

1. Experiment Purpose

Experiment 19 moves Logos from training-side scaling to deployment-side efficiency. The goal was to take the best trained model from Experiment 18 and test whether the model can be stored and used more efficiently without retraining.

The experiment compares the original FP32-style reference checkpoint, an FP16 checkpoint, and a dynamic INT8 CPU variant. The main measurements are checkpoint size, estimated validation loss, estimated perplexity, generation speed, latency, and memory use.

2. Source Model

Property	Value
Source experiment	Experiment 18: Large Scaling + 15k Training
Architecture	Decoder-only GPT-style Transformer
Parameters	124,031,232
Dataset	OpenWebText 100k samples
Tokenizer	GPT-2 tokenizer
Context length	512 tokens
Layers / Embedding / Heads	12 layers / 768 embedding dim / 12 heads
FFN type	SwiGLU
Weight tying	Enabled
Experiment 18 final validation loss	3.9811
Experiment 18 final perplexity	53.57
Original checkpoint size	473.20 MB

3. Compression Variants Tested

Variant	Purpose	Device
FP32 reference	Reference model copied from the Experiment 18 checkpoint. Used as the baseline for comparison.	CUDA
FP16 checkpoint	Floating-point tensors are stored in half precision to reduce checkpoint size and GPU memory use.	CUDA
Dynamic INT8 CPU	Dynamic quantization is applied mainly to Linear layers. This is a CPU-side deployment compression test.	CPU

4. Results

Variant	Device	Checkpoint	Val loss est.	PPL est.	Gen speed	Latency	Peak VRAM
FP32 reference	CUDA	473.20 MB	3.9931	54.22	105.55 tok/s	9.47 ms/token	6518.82 MB
FP16 checkpoint	CUDA	310.24 MB	3.9297	50.89	109.38 tok/s	9.14 ms/token	5958.36 MB
Dynamic INT8 CPU	CPU	267.05 MB	3.9660	52.77	49.16 tok/s	20.34 ms/token	N/A

5. Key Observations

FP16 is the best practical GPU deployment variant. It reduced checkpoint size from 473.20 MB to 310.24 MB, while keeping estimated validation quality close to the FP32 reference. It also slightly improved measured generation latency in this run.

Dynamic INT8 CPU produced the smallest saved state. The INT8 CPU state was 267.05 MB, smaller than both FP32 and FP16. However, generation latency was worse because the test ran on CPU rather than CUDA.

Quality differences should be interpreted carefully. The validation estimates were computed on sampled batches. Therefore, FP16 appearing slightly better than FP32 is likely measurement noise rather than a real quality improvement.

6. Scientific Conclusion

Experiment 19 is a successful deployment-side efficiency test. It shows that the best Logos model from Experiment 18 can be compressed substantially while maintaining approximately similar language-modeling quality on sampled validation batches.

The safest conclusion is that FP16 provides the strongest GPU deployment tradeoff: it reduces checkpoint size by about 34.4%, lowers measured VRAM use, and gives similar or slightly better observed inference speed without a meaningful quality drop.

Dynamic INT8 is useful as a CPU memory-reduction experiment, but it is not the best speed configuration in this setup. Its main value is showing that CPU-side model storage can be reduced further, though at the cost of higher generation latency.

7. Relation to the Logos Goal

The Logos goal is to improve language models not only by increasing raw capability, but by improving capability per parameter, memory byte, latency unit, checkpoint size, and watt. Experiment 19 directly supports this goal because it evaluates the same trained model under different deployment representations.

Earlier experiments improved model quality through architecture choices, scaling, and longer training. Experiment 19 extends the project to post-training efficiency by showing that the best model can be compressed for more practical deployment.

8. Reporting Warning

Do not claim that FP16 improves model intelligence. The apparent validation-loss improvement is based on sampled-batch evaluation and should be treated as noise. The correct claim is that FP16 preserves approximate validation quality while reducing checkpoint size and memory use.

Also, dynamic INT8 quantization in this notebook only targets specific module types, mainly Linear layers. It does not represent a fully optimized production-grade GPU INT8 or 4-bit LLM deployment pipeline.

9. Recommended Next Steps

Step	Purpose
Create a Logos Efficiency Index	Turn raw metrics into a single resource-aware ranking score.
Run full or more stable validation estimates	Reduce noise in validation comparisons across FP32, FP16, and INT8.
Test GPU-oriented low-bit deployment later	Consider 8-bit/4-bit deployment using libraries built for LLM inference if available.
Update the final paper/report	Include Experiment 19 as the deployment-side efficiency section.

10. One-Line Summary

Experiment 19 showed that the 124M-parameter Logos model from Experiment 18 can be compressed for deployment, with FP16 giving the best practical GPU tradeoff by reducing checkpoint size from 473.20 MB to 310.24 MB while preserving approximate validation quality.